

Columbia Econ Math Camp 2025

Multivariate Calculus

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August 11, 2025

1. Derivatives (*)

Univariate Derivatives*

Definition 1 (Univariate Differentiability)

Let open $A \subset \mathbb{R}$, $x \in A$. $f : A \rightarrow \mathbb{R}$ is **differentiable** at x iff

$$f'(x) := \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists, and if so, the function $f' : A \rightarrow \mathbb{R}$ is called its **derivative (function)**.

Exercise 1

In what follows, assume $f : [-1, 1] \rightarrow \mathbb{R}$

1. Show that $f(x) = 1(x > 0)$ does not have a derivative at 0.
2. Show that if f is differentiable then it must be continuous.

Univariate Derivatives*

Notable results for derivatives

- Useful derivatives: $(x^\alpha)' = \alpha x^{\alpha-1}$ ($\alpha \neq 0$); $(\ln x)' = 1/x$; $(e^x)' = e^x$; $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$
- $(f + g)' = f' + g'$ where $f, g : A \rightarrow \mathbb{R}$.
- $(\lambda \cdot f)' = \lambda \cdot f'$ where $\lambda \in \mathbb{R}$ (Scalar product rule).
- $(fg)' = f'g + fg'$ (Product rule) - Try to prove this at home!
- $(f/g)' = (f'g - fg')/g^2$ (Division rule).

Theorem 1 (L'Hôpital's Rule)

Let open $A \subset \mathbb{R}$, $a \in A$, $f, g : A \rightarrow \mathbb{R}$ be differentiable everywhere except possibly at a itself. Suppose that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0 \quad \text{or} \quad \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \pm\infty,$$

and that $g'(x) \neq 0$ for all $x \in I \setminus \{a\}$. If the limit

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

exists (or is $\pm\infty$), the

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Univariate Derivatives*

Theorem 2 (Mean Value Theorem or MVT)

Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. Then there exists $x \in (a, b)$ such that

$$f'(x) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 3 (Intermediate Value Theorem or IVT)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$, and $u \in (f(a), f(b))$. Then there exists $c \in (a, b)$ such that $u = f(c)$

Exercise 2

For a differentiable function $f : [a, b] \rightarrow \mathbb{R}$, show (at home) that if $f' > 0$ on (a, b) then f is increasing on (a, b) . Hint: Use MVT.

Multivariate Derivatives*

Denote the j -th **canonical vector** in $\mathbb{R}^n$ as $\mathbf{e}_j = (0, 0, \dots, 1, \dots, 0)$, where 1 is in the j -th element.

Definition 2 (Partial Derivative)

Let open $A \subset \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a (multivariate) function, and let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in A$, so $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$. The **partial derivative** of f_i with respect to the i -th argument at the point $\mathbf{x}$, provided it exists, is defined as

$$\frac{\partial f_i}{\partial x_j}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f_i(\mathbf{x} + h\mathbf{e}_j) - f_i(\mathbf{x})}{h},$$

More commonly you might know the partial by this equivalent definition:

$$\frac{\partial f_i}{\partial x_j}(\mathbf{x}) = \left. \frac{d}{dh} f_i(\mathbf{x} + h\mathbf{e}_j) \right|_{h=0}$$

E.g. if $f(x, y) = x^2 + y^2$ then $\frac{\partial f}{\partial x}(x, y) = \left. \frac{d}{dh} (x + h)^2 + y^2 \right|_{h=0} = 2x$

Multivariate Derivatives*

For a $\mathbb{R}^n$ to $\mathbb{R}^m$ function, we can collect all the mn partial derivatives in the so-called Jacobian matrix is an $m \times n$ matrix defined as follows

Definition 3 (Jacobian Matrix)

Let open $A \subset \mathbb{R}^n$ and $\mathbf{x} \in A$. If a function $f : A \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{x}$, then the Jacobian Matrix is defined

$$J_f(\mathbf{x}) := \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \frac{\partial f_1}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \frac{\partial f_2}{\partial x_1}(\mathbf{x}) & \frac{\partial f_2}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_2}{\partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}) & \frac{\partial f_m}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}) \end{bmatrix}$$

When $m = 1$, the Jacobian becomes the **gradient** of f - by convention it is a row vector - denoted as

$$\nabla f(\mathbf{x}) := \left[\frac{\partial f}{\partial x_1}(\mathbf{x}) \quad \frac{\partial f}{\partial x_2}(\mathbf{x}) \quad \cdots \quad \frac{\partial f}{\partial x_n}(\mathbf{x}) \right]$$

Multivariate Derivatives*

Definition 4 (Total Differentiability)

Let open $A \subset \mathbb{R}^n$ and $\mathbf{x} \in A$. A function $f : A \rightarrow \mathbb{R}^m$ is said to be **(totally) differentiable** at $\mathbf{x}$ iff there exists an $m \times n$ real matrix $\mathbf{C}$ such that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \mathbf{C}\mathbf{h}\|}{\|\mathbf{h}\|} = 0.$$

In this case, define the **(total) derivative** of f at $\mathbf{x}$ as the matrix $\mathbf{C}$. More commonly the total derivative is denoted as $f'(\mathbf{x})$ or $Df(\mathbf{x})$.

Here $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$.

- If it exists, $\mathbf{C}$ is unique (trust me).
- Intuitively, if $\mathbf{h} = d\mathbf{x} \rightarrow \mathbf{0}$, the definition is saying that

$$df(\mathbf{x}) = f(\mathbf{x} + d\mathbf{x}) - f(\mathbf{x}) = Df(\mathbf{x}) \cdot d\mathbf{x}$$

This is known informally as **taking the total derivative** of f .

Multivariate Derivatives*

In the case of real-valued functions, like above, you can show that if f' exists, then it must be equal to the Jacobian matrix of f at $\mathbf{x}$.

Theorem 4 (The Jacobian Matrix Theorem)

Let open $A \subset \mathbb{R}^n$ and $\mathbf{x} \in A$. If a function $f : A \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{x}$, then $\frac{\partial f_i}{\partial x_j}(\mathbf{x})$ exists for any $(i, j) \in \{1, \dots, m\} \times \{1, \dots, n\}$, and furthermore we have

$$f'(\mathbf{x}) = J_f(\mathbf{x})$$

Exercise 3

Prove the Jacobian Matrix Theorem (at home). Hint: Recall that if $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ then $|x_i| < \|\mathbf{x}\|$ for every $i \in n$

Note that the reverse may not hold! (i.e. all partials may exist but f is not differentiable).

Multivariate Derivatives*

Theorem 5 (Chain Rule)

Let $S \subset \mathbb{R}^n$, $\mathbf{x} \in \text{int}(S)$, and $f : S \rightarrow \mathbb{R}^m$. Let T be such that $f(S) \subset T \subset \mathbb{R}^m$ and $f(\mathbf{x}) \in \text{int}(T)$, and let $g : T \rightarrow \mathbb{R}^k$. If f is differentiable at $\mathbf{x}$, and g is differentiable at $f(\mathbf{x})$, then $g \circ f : S \rightarrow \mathbb{R}^k$ is differentiable at $\mathbf{x}$. Furthermore, we have:

$$(g \circ f)'(\mathbf{x}) = g'(f(\mathbf{x})) \cdot f'(\mathbf{x})$$

- These derivatives are all matrices of following dimension: $g'(f(\mathbf{x}))$ is $m \times k$, $f'(\mathbf{x})$ is $n \times m \rightarrow (g \circ f)'(\mathbf{x})$ is $n \times k$.
- To visualize better, note that

$$\frac{\partial (g \circ f)_i}{\partial x_j}(\mathbf{x}) = \sum_{l=1}^m \left[\frac{\partial g_i}{\partial y_l}(f(\mathbf{x})) \cdot \frac{\partial f_l}{\partial x_j}(\mathbf{x}) \right]$$

Intuitively, changing x_j perturbs all m elements of f according to its partials (∂f_l) , which in turn perturbs g_i through its own partials (∂g_i) .

2. Implicit Functions

Implicit Functions

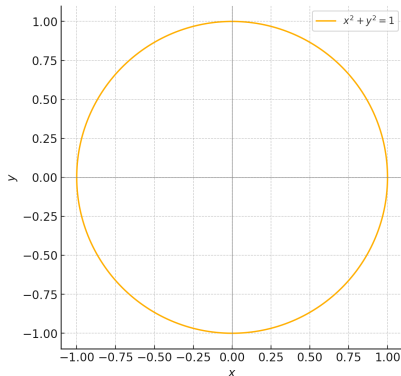
Definition 5 (Implicit Function)

Let $\mathbf{y} \in \mathbb{R}^m$, $\mathbf{x} \in \mathbb{R}^n$ and function $H : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^m$ such that $H(\mathbf{x}, \mathbf{y}) = \mathbf{0}$. In this case, we say that H defines an **implicit relation or function** between $\mathbf{x}$ and $\mathbf{y}$.

- Any equilibrium of a static model can be written in the form $f(\mathbf{x}, \mathbf{y}) = \mathbf{0}$, as we will learn later.
- An **(explicit) function** of H is a transformation $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that $\mathbf{y} = f(\mathbf{x})$ and $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$.
 - ▶ Often in applications we care a lot that such g exists.
- In many cases, there does not always exist an explicit function that describes the entire implicit relation.

Implicit Functions

- Graphically, an implicit function is an “obvious” relationship between two groups of variables.
- However, in general this relationship may not have a correspondence rule $y = g(x)$ (i.e. you can’t “solve” for y).



Implicit Functions

The following theorem illustrates conditions and a formula for when we can *locally* express an implicit function as an explicit function.

Theorem 6 (Implicit Function Theorem)

Let f be a function from $A \subset \mathbb{R}^n \times \mathbb{R}^m$ to $\mathbb{R}^m$. Let $(\mathbf{x}_0, \mathbf{y}_0) \in \text{int}(A)$ such that $f(\mathbf{x}_0, \mathbf{y}_0) = 0$. If f is C^1 at $(\mathbf{x}_0, \mathbf{y}_0)$ and the $m \times m$ Jacobian matrix $f'_y(\mathbf{x}_0, \mathbf{y}_0)$ is invertible, then there exist an open ball B_x around $\mathbf{x}_0$ and an open ball B_y around $\mathbf{y}_0$ such that for all $\mathbf{x} \in B_x$, there exists a unique $\mathbf{y} \in B_y$ such that $f(\mathbf{x}, \mathbf{y}) = 0$. Therefore, the equation $f(\mathbf{x}, \mathbf{y}) = 0$ implicitly defines a function $g : B_x \rightarrow B_y$ with the property

$$f(\mathbf{x}, g(\mathbf{x})) = 0$$

for any $\mathbf{x} \in B_x$. Further, g is unique, differentiable at any $\mathbf{x} \in B_x$, and

$$g'(\mathbf{x}) = - \left[f'_y(\mathbf{x}, g(\mathbf{x})) \right]^{-1} f'_x(\mathbf{x}, g(\mathbf{x})) \quad (1)$$

Implicit Functions

- Notation: The term f'_x [resp. f'_y] refers to the submatrix of f' that holds the partials of f wrt $\mathbf{x}$ [resp. $\mathbf{y}$], thus

$$f'(\mathbf{x}_0, \mathbf{y}_0) = [f'_x(\mathbf{x}_0, \mathbf{y}_0) \quad f'_y(\mathbf{x}_0, \mathbf{y}_0)]$$

- The formula is simple to obtain from the Theorem. Since $f(\mathbf{x}, g(\mathbf{x})) = \mathbf{0}$ around $\mathbf{x}$, we apply the derivative wrt $\mathbf{x}$ on both sides

$$f'_x(\mathbf{x}, g(\mathbf{x})) + f'_y(\mathbf{x}, g(\mathbf{x})) \cdot g'(\mathbf{x}) = \mathbf{0}$$

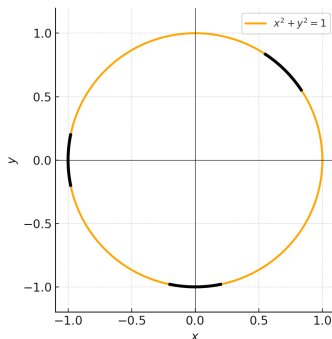
Then rearranging and using $\mathbf{y} = g(\mathbf{x})$ we get

$$g'(\mathbf{x}) = - \left[f'_y(\mathbf{x}, g(\mathbf{x})) \right]^{-1} f'_x(\mathbf{x}, g(\mathbf{x}))$$

- The hard part is understanding how can $f'_y(\mathbf{x}_0, \mathbf{y}_0)$ is invertible ensure that $f'_y(\mathbf{x}, \mathbf{y})$ is invertible in any $\mathbf{x} \in B_x$. The next slide illustrates this

Implicit Functions

- In $\mathbb{R}^2$, the theorem tells us that we can find g such that $f(x, g(x)) = 0$ around (x_0, y_0) as long as $\partial f / \partial y(x_0, y_0) \neq 0$; this can be seen clearly comparing the three black arcs below:



- The intuition here is that there is an indeterminacy at point $(-1, 0)$: When x changes, we can't unequivocally infer the change in y . The relationship between x, y around that point is not injective.

3. Taylor Expansions

Higher Order Derivatives

Definition 6 (C^k functions)

Let $A \subset \mathbb{R}^n$ $f : A \rightarrow \mathbb{R}^m$. If f' is differentiable at $x \in \text{int}(A_1)$, we call the derivative of f' at x , the **second derivative** of f at x , and denote it as $f''(x)$. In this case, we say that f is **twice differentiable** at x . If f'' is differentiable at $x \in \text{int}(A_2)$, we call the derivative of f'' at x , an $m \times n^2 \times n$ real matrix, the **third derivative** of f at x , and denote it as $f'''(x)$. In this case, we say that f is **three times differentiable** at x . Inductively, we can define the k -th order derivative of f at x , an $m \times n^{k-1} \times n$ real matrix, and denote it as $f^{(k)}(x)$.

We say that f from $A \subset \mathbb{R}^n$ to $\mathbb{R}^m$ is **k -th continuously differentiable** at x iff $x \in \text{int}(A_k)$ and $f^{(k)}(x)$ is continuous as a function, where A_k is the set of points at which $f^{(k-1)}$ is differentiable. In this case, f is said to be C^k at x . We say that f is **k -th continuously differentiable** if A is open and f is k -th continuously differentiable at all $x \in A$. In this case, f is said to be C^k .

Higher Order Derivatives

Now let's define the notion of the second order derivative for multivariate functions:

Definition 7 (Hessian Matrix)

Let open $A \subset \mathbb{R}^n$ $f : A \rightarrow \mathbb{R}$ be twice continuously differentiable (or C^2). The **Hessian matrix** of f at $\mathbf{x} \in \text{int}(A)$, denoted by $H_f(\mathbf{x})$, is the $n \times n$ matrix of second-order partial derivatives of f at $\mathbf{x}$:

$$H_f(\mathbf{x}) = (\nabla f)'(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}$$

Higher Order Derivatives

As you all probably know, in most functions you will deal with the cross-partials are symmetric:

Theorem 7 (Young's Theorem)

In the setting above, when f is C^2 at $\mathbf{x}$, then all cross partials exist at $\mathbf{x}$, and further, $H_f(\mathbf{x})$ is symmetric.

Discontinuities can break Young's Theorem, even if cross partials exist:

Exercise 4

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{otherwise} \end{cases}$$

Show that $\frac{\partial^2 f}{\partial y \partial x}(0, 0) \neq \frac{\partial^2 f}{\partial x \partial y}$.

Taylor Expansions

It turns out that continuous differentiable functions can be very accurately expressed as some polynomial or power series. Let's define an expansion:

Definition 8 (Taylor Expansion)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be C^n . Then the **n -th order Taylor polynomial** of f at a is:

$$P_{f,n}(t) := f(a) + f'(a)(t-a) + \frac{f''(a)(t-a)^2}{2} + \dots + \frac{f^{(n)}(a)(t-a)^n}{n!}$$

When $n \rightarrow \infty$ then we have the **Taylor series** of f at a :

$$P_{f,\infty}(t) := f(a) + f'(a)(t-a) + \frac{f''(a)(t-a)^2}{2} + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(t-a)^n}{n!}$$

Usually when $a = 0$ these are called the **MacLaurin series** of f .

Taylor Expansions

Taylor expansions are immensely popular because these polynomials can be used to approximate a wide class of function with the following result:

Theorem 8 (Univariate Taylor theorem)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be C^n , $a \in \mathbb{R}$, $h > 0$. Then $\exists \zeta \in (a, a + h)$ s.t.

$$f(a + h) = p_{n-1}(a + h) + \frac{f^{(n)}(\zeta)h^n}{n!}$$

Here $p_{n-1}(a + h) = \sum_{i=0}^{n-1} \frac{f^{(i)}(a)(h-a)^i}{i!}$ is a Taylor polynomial of f at a and term $R_n(h) := \frac{f^{(n)}(\zeta)h^n}{n!}$ is called the **Taylor remainder**, which further satisfies $R(h) = o(|h|)$ (i.e. $R_n(h) \rightarrow 0$ as $h \rightarrow 0$).

In words, we can **approximate** f using p_{n-1} (or $f \approx p_{n-1}$) and the approximation will get better as $h \rightarrow 0$ (as $R_n(h) \rightarrow 0$) or $n \rightarrow \infty$ (because $n!$ grows faster than either $f^{(n)}$ or h^n). Note this Theorem is a generalization of the MVT.

Taylor Expansions

In fact, in many cases, function f is actually equal to its Taylor series over some interval with non-zero measure (i.e. not just some tiny open ball).

Definition 9 (Analytical functions)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be **analytical** if for some $a \in \mathbb{R}$ and $D \subset \mathbb{R}$, we can define $f_D(x) = p_{f,\infty}(x)$ for any $x \in D$, where $p_{f,\infty}$ is the Taylor series of f at a .

- The subset $D := (a - R, a + R)$ is known as the **interval of convergence** for f , and thus R is called the **radius of convergence**.
- One can show that:

$$R = \lim_{k \rightarrow \infty} \left| \frac{f^{(k)}(a)}{f^{(k+1)}(a)} (k+1) \right|$$

Intuitively, this results from solving an inequality that ensures that as $k \rightarrow \infty$, the terms of the Taylor series are shrinking in size.

Taylor Expansions

Instead of defining conditions for when a function is analytical, let me just list for you the MacLaurin series for some canonical functions:

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}$
- $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad x \in \mathbb{R}$
- $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad x \in \mathbb{R}$
- $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$ (in our notation, convergence radius is $R = 1$).
- $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}, \quad |x| < 1$
- $(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \quad |x| < 1, \alpha \in \mathbb{R}$

Taylor Expansions

You can use these canonical forms and the following result to compute the MacLaurin series of pretty much anything:

Proposition 1

Let $F = f \circ g$, where $f, g : \mathbb{R} \rightarrow \mathbb{R}$. Then the Taylor series for F is

$$p_{F,\infty}(a+h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(h)}{n!} (g(h))^n$$

So for example $e^{-\frac{x^2}{2}} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-x^2}{2} \right)^n$

Taylor Expansions

Finally, let me write down the version of the Taylor theorem for multivariate functions:

Theorem 9 (Multivariate Taylor theorem)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 at $\mathbf{x} \in \mathbb{R}^n$. Then

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|)$$

If f is C^3 at $\mathbf{x}$, we have

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x})\mathbf{h} + \frac{1}{2}\mathbf{h}^T H_f(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|^2)$$

4. Log-linearization

Log-linearization

- A key result in approximation is that percentual changes from some scalar y^* to y is approximately equal to its own log-deviation, i.e $\Delta\%y \approx \ln(y) - \ln(y^*)$.
- To see why, note that for infinitesimal change we have:

$$\Delta\%y = \frac{y - y^*}{y^*} \approx \frac{dy^*}{y^*} = d \ln(y^*) \approx \ln\left(\frac{y}{y^*}\right)$$

- This is similar for percentual changes of functions:

$$\Delta\%f(x^*) = \frac{f(x) - f(x^*)}{f(x^*)} \approx \frac{df(x^*)}{f(x^*)} = d \ln f(x^*) \approx \ln(f(x)) - \ln(f(x^*))$$

- For vectors in $\mathbb{R}^n$, this extends element wise as follows:
 $\widehat{\mathbf{y}^*} := \ln(\mathbf{y}/\mathbf{y}^*) \approx d\mathbf{y}^*/\mathbf{y}^*$ and $\widehat{f(\mathbf{x})} = \ln(f(\mathbf{x})/f(\mathbf{x}^*)) \approx \frac{df(\mathbf{x}^*)}{f(\mathbf{x}^*)}$.
- Further let's define the **elasticity of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ w.r.t x_i at $\mathbf{x}^*$** as
 $\eta_i(\mathbf{x}^*) := \frac{f_i(\mathbf{x}^*)x_i^*}{f(\mathbf{x}^*)}$

Log-linearization

- Log-linearizing any real function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is simply an application of the total differential of a function wrt a point $\mathbf{x}^*$.
- Start by taking the total differential of f at $\mathbf{x}^*$.

$$df(\mathbf{x}^*) \approx f_1(\mathbf{x}^*)dx_1^* + f_2(\mathbf{x}^*)dx_2^* + \dots + f_n(\mathbf{x}^*)dx_n^*$$

- Divide both sides by $f(\mathbf{x}^*)$, and in the RHS, for each term i divide and multiply x_i^* :

$$\frac{df(\mathbf{x}^*)}{f(\mathbf{x}^*)} \approx \frac{f_1(\mathbf{x}^*)x_1^*}{f(\mathbf{x}^*)} \frac{dx_1^*}{x_1^*} + \frac{f_2(\mathbf{x}^*)x_2^*}{f(\mathbf{x}^*)} \frac{dx_2^*}{x_2^*} + \dots + \frac{f_n(\mathbf{x}^*)x_n^*}{f(\mathbf{x}^*)} \frac{dx_n^*}{x_n^*}$$

- Plugging in the results from last slide, we get the main result from this topic:

$$\widehat{f(\mathbf{x}^*)} \approx \eta_1(\mathbf{x}^*)\widehat{x_1^*} + \eta_2(\mathbf{x}^*)\widehat{x_2^*} + \dots \eta_n(\mathbf{x}^*)\widehat{x_n^*}$$

Proposition 2

Some quick but useful results (α, c are scalar parameters, x and x_i are scalar variables that vary across time):

- $\widehat{x_1 x_2} = \widehat{x_1} + \widehat{x_2}$, $\widehat{cx} = \widehat{x}$ and $\widehat{c} = 0$.
- $\widehat{x^\alpha} = \alpha \widehat{x}$. Combined with above, we get $\widehat{x_1/x_2} = \widehat{x_1} - \widehat{x_2}$.
- $\widehat{x_1 + x_2} = \frac{x_1^*}{x_1^* + x_2^*} \widehat{x_1} + \frac{x_2^*}{x_1^* + x_2^*} \widehat{x_2}$
- $\widehat{t(x)} = \frac{t'(x^*)x^*}{t(x^*)} \widehat{x} = \eta_x(x^*) \widehat{x}$

- Note that you can talk about $\widehat{y}$ and $\widehat{y^*}$ interchangeably.
- Finally, if you have an equation in your model such as $f(x) = g(x)$, then clearly $\widehat{f(x)} = \widehat{g(x)}$, so you can log-linearize them separately and then equate them.

5. Homogeneous Functions

Homogeneous Functions

We will introduce the concept of homogeneous functions, which appear everywhere in economic theory. First let's define what a cone is

Definition 10 (Cone)

A set C in real vector space V is said to be a **cone** iff for any scalar $\lambda > 0$ and vector $\mathbf{v} \in C$, we have $\lambda \mathbf{v} \in C$.

- As its name suggests, visually they have the appearance of (infinitely large) cones.
- Any quadrant of $\mathbb{R}^n$ is a cone, such as $\mathbb{R}_{++}^n$, but also all of $\mathbb{R}^n$ is a cone as well.
- These will be important so that homogeneous functions are well-defined.

Homogeneous Functions

Definition 11 (Homogeneous functions)

Let C be a cone in real vector space V , and let W be another real vector space. For $k \in \mathbb{R}$, we say $f : C \rightarrow W$ is **homogeneous of degree k** iff for any scalar $\lambda > 0$ and vector $\mathbf{v} \in C$, we have $f(\lambda \mathbf{v}) = \lambda^k f(\mathbf{v})$.

- For most applications $C = \mathbb{R}_+^n$ and $W = \mathbb{R}$.
- Production functions are usually homogeneous of some degree (e.g. Cobb Douglas, CES, etc).

Homogeneous Functions

A nice property is that cross partials of homogenous functions are homogenous themselves, but at one degree less.

Proposition 3

Let C be a cone in $\mathbb{R}^n$, $f : C \rightarrow \mathbb{R}$ be homogeneous of degree k , $\mathbf{x} \in \text{int}(C)$, $\lambda > 0$. If $f_i := \frac{\partial f}{\partial x_i}$ exists at $\mathbf{x}$, then it also exists at $\lambda \mathbf{x}$ and further we have $f_i(\lambda \mathbf{x}) = \lambda^{k-1} f_i(\mathbf{x})$.

With this proposition we can prove the famous Euler Equation:

Theorem 10 (Euler's equation)

Let C be a cone in $\mathbb{R}^n$, $f : C \rightarrow \mathbb{R}$ be homogeneous of degree k differentiable at $\mathbf{x} \in \text{int}(C)$. Then, denoting the dot product as $\cdot$, we have that

$$\nabla f(\mathbf{x}) \cdot \mathbf{x} = kf(\mathbf{x})$$

Homothetic Functions**

Definition 12 (Homothetic functions)

Let C be a cone in real vector space V . We say $f : C \rightarrow \mathbb{R}$ is **homothetic** iff there exists $h : C \rightarrow \mathbb{R}$ that is homogeneous of some degree k and $g : \mathbb{R} \rightarrow \mathbb{R}$ that is strictly increasing s.t. $f = g \circ h$.

The useful property here is that $f(\mathbf{u}) = f(\mathbf{w})$ implies $f(\lambda \mathbf{u}) = f(\lambda \mathbf{w})$ for $\mathbf{u}, \mathbf{w} \in C$ and $\lambda > 0$. Further this implies a famous result of homothetic functions in economics.

Proposition 4

Let C be a cone in $\mathbb{R}^n$, $f : C \rightarrow \mathbb{R}$ is homothetic, $\mathbf{x} \in \text{int}(C)$, $\lambda > 0$. If f is differentiable at $\mathbf{x}$ and $\lambda \mathbf{x}$, and $f_j(\lambda \mathbf{x})$ and $f_j(\mathbf{x})$ are not zero then

$$\frac{f_i(\lambda \mathbf{x})}{f_j(\lambda \mathbf{x})} = \frac{f_i(\mathbf{x})}{f_j(\mathbf{x})}$$

6. Integrals*

Integrals*

Let me show you (wlog) the simplest definition of integrability I know of:

Definition 13 ((Simplified) Riemann Integrability)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Let's partition $[a, b]$ into n equally spaced intervals $\{C_i\}_{i=1}^n$ - i.e. $\bigcup_{i=1}^n C_i = [a, b]$ with $C_i \cap C_j = \emptyset$ for any $i \neq j$. The **lower** and **upper Riemann sums** are defined:

$$U(f) = \frac{b-a}{n} \sum_{i=1}^n \sup_{x_i \in C_i} f(x_i), \quad L(f) = \frac{1}{n} \sum_{i=1}^n \inf_{x_i \in C_i} f(x_i),$$

The function f is Riemann integrable if the lower and upper Riemann sums converge to the same number as $n \rightarrow \infty$. This value is called the **Riemann integral** of f over $[a, b]$, and is denoted by:

$$\int_a^b f(x) dx := \lim_{n \rightarrow \infty} U(f) = \lim_{n \rightarrow \infty} L(f)$$

Integrals*

Note that each sum term in both Riemann sum is just the area of a rectangle where

- Base is the sub-interval length (which is the same for all intervals by construction).
- Height is the highest (for the upper sum) and lowest (for lower sum) value f achieves in each interval C_i .

As $n \rightarrow \infty$ and under some conditions, both sums are equal and, graphically, are equivalent to the **area under the curve** of f between points a and b . For example:

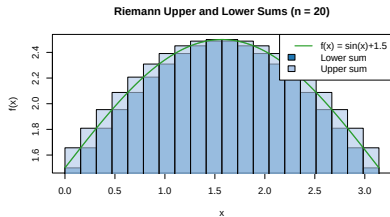
Theorem 11 (Continuity and Integrability)

Let f be a bounded real function on $[a, b]$ which is discontinuous only at finitely many points on $[a, b]$. Then f is Riemann integrable on $[a, b]$.

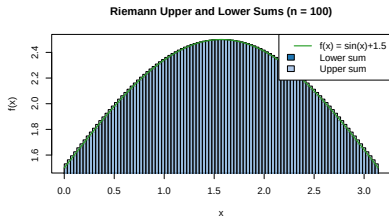
Integrals*

A visualization of integrability

(a) $n=20$



(b) $n=100$



- You can calculate the integral of any (integrable) function by evenly partitioning the domain into very small cells, computing the maximum (or minimum/midpoint) of the function in each cells, and summing the cells' areas/volumes.
- If I don't have a computer to do the above, how can we **in practical** terms calculate this integral?
- The Fundamental Theorem of Calculus gives a very simple answer.

Definition 14 (Antiderivative)

Let $f : [a, b] \rightarrow \mathbb{R}$. We say $F : [a, b] \rightarrow \mathbb{R}$ is an **antiderivative** of f on $[a, b]$ if

$$F'(x) = f(x) \quad \text{for all } x \in [a, b].$$

Antiderivatives of some commonly used functions:

- $\int x^\alpha dx = \frac{1}{\alpha+1}x^{\alpha+1} + C, \quad \alpha \neq -1$
- $\int \frac{1}{x} dx = \ln(|x|) + C, \quad x \neq 0$
- $\int e^{\alpha x} dx = \frac{1}{\alpha}e^{\alpha x} + C, \quad \alpha \neq 0$
- $\int \sin x dx = -\cos x + C$

where $C \in \mathbb{R}$ is a constant.

Theorem 12 (Fundamental Theorem of Calculus)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function with antiderivative $F : [a, b] \rightarrow \mathbb{R}$ (i.e. $F'=f$). Then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Another useful property is that, like derivatives, integral is a **linear operator**:

- $\int_a^b (f + g)(x)dx = \int_a^b f(x)dx + \int_a^b g(x)dx$
- $\int_a^b (\lambda \cdot f)(x)dx = \lambda \cdot \int_a^b f(x)dx$, where $\lambda \in \mathbb{R}$

Also, if $\{C_i\}_{i=1}^n$ is a partition of A , then

$$\int_A f(x)dx = \sum_{i=1}^n \int_{C_i} f(x)dx$$

Integrals*

Three useful tricks to solve integrals

1. Differentiation of α :

$$\int_a^b f(x) d\alpha(x) = \int_a^b f(x) \alpha'(x) dx$$

2. Change of variable:

$$\int_a^b f(\phi(x)) d\alpha(\phi(x)) = \int_{\phi(a)}^{\phi(b)} f(y) d\alpha(y)$$

3. Integration by parts:

$$\int_a^b f(x) dg(x) = f(b)g(b) - f(a)g(a) - \int_a^b g(x) df(x)$$

Exercise 5

Prove the integration by parts property (at home). Hint: Use the product rule of derivatives.

Integrals*

Consider a function $I(x) = \int_{u(x)}^{v(x)} f(x, t) dt$. How do we compute the derivative I' ? The following theorem gives us conditions and very simple formula to do so:

Theorem 13 (Leibniz's Formula)

Let, $A \subset \mathbb{R}^2$, $f : A \rightarrow \mathbb{R}$, $E := [a, b] \times [c, d] \subset A$. Consider two continuously differentiable functions $u, v : [a, b] \rightarrow [c, d]$. If $\frac{\partial f}{\partial x}(x, t)$ exists for all $(x, t) \in E$ and is continuous on E , then the function

$$I(x) := \int_{u(x)}^{v(x)} f(x, t) dt$$

is differentiable on $[a, b]$, and

$$I'(x) = f(x, v(x))v'(x) - f(x, u(x))u'(x) + \int_{u(x)}^{v(x)} \frac{\partial f}{\partial x}(x, t) dt.$$

Multiple Integrals**

Theorem 14 (Fubini's Theorem)

Let $C_X = [a_1, b_1] \times \cdots \times [a_k, b_k]$ and $C_Y = [a_{k+1}, b_{k+1}] \times \cdots \times [a_{k+m}, b_{k+m}]$. For continuous $f : C_X \times C_Y \rightarrow \mathbb{R}$, we have

$$\begin{aligned} \int_{C_X \times C_Y} f(x, y) d(x, y) &= \int_{C_Y} \left(\int_{C_X} f(x, y) dx \right) dy \\ &= \int_{C_X} \left(\int_{C_Y} f(x, y) dy \right) dx \end{aligned}$$

If $C = [a_1, b_1] \times \cdots \times [a_k, b_k]$ and $f : C \rightarrow \mathbb{R}$ is continuous, then we can iterate Fubini's theorem to find,

$$\begin{aligned} \int_C f(x) dx &= \int_{[a_1, b_1] \times \cdots \times [a_k, b_k]} f(x_1, \dots, x_k) dx \\ &= \int_{a_k}^{b_k} \left(\int_{a_{k-1}}^{b_{k-1}} \cdots \left(\int_{a_2}^{b_2} \left(\int_{a_1}^{b_1} f(x_1, \dots, x_k) dx_1 \right) dx_2 \cdots \right) dx_{k-1} \right) \end{aligned}$$

Multiple Integrals**

However, if the integration region is parametrized somehow, Fubini does not hold. Suppose we want to integrate continuous $f(x, y)$ over a set

$$A := \{(x, y) \in \mathbb{R}^2 : x \in [a, b], y \in [u(x), v(x)]\}.$$

Then this can only be done as follows

$$\int_A f(x, y) \, dx dy = \int_a^b \left(\int_{u(x)}^{v(x)} f(x, y) \, dy \right) dx$$

Multiple Integrals**

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Multiple Integrals**

Next we show **change of variables in double integrals**:

- Consider double integral $\int_A f(x, y) dx dy$. Suppose that you want to implement the change of variables

$$x = g(u, v), \quad y = h(u, v)$$

Here the change defines a one-to-one C^1 transformation from set A' in the uv -plane onto set A in xy -plane.

- Assume A, A' are bounded and open, f is continuous and bounded on A .
- Then the change can be done through the following formula:

$$\int_A f(x, y) dx dy = \int_{A'} f(g(u, v), h(u, v)) \left| \frac{\partial(g, h)}{\partial(u, v)} \right| du dv$$

where the partial inside the absolute value must be bounded on A' and is

$$\frac{\partial(g, h)}{\partial(u, v)} := \det \begin{pmatrix} \partial g / \partial u & \partial g / \partial v \\ \partial h / \partial u & \partial h / \partial v \end{pmatrix}$$